Sun-Proof Cover – Three Options Overview

Option	Material & Process	Key Components
Option 1	SUS304 / SUS316 stainless steel, 3mm thick; Laser cut → Bending → Chamfer → Powder coated	E-ring&washers &Ø10mm E-ring shaft – all SUS304 standard parts (frequent friction, has painting fall off risk)
Option 2	Aluminum, 8mm thick; CNC → Anodizing → Powder coated	SUS304 PWM4X12 screws, Al washers, Ø10mm Al shaft (all Al parts: CNC → Anodizing → Powder coated, has fall off risk)
Option 3	PVDF, 8mm thick; CNC	4mm PVDF washer, Ø10mm PVDF shaft, all CNC to make.

We suggest option 3 that use PVDF material for chlorine gas protection.

Sun-Proof Cover – Three Options Overview

The sun-proof cover can be opened freely without sliding it far forward, allowing access to add salt.

Comparison of the three options:
Option 1: SUS304 / SUS316 offers sufficient structural strength and good weather resistance, but is at risk of corrosion when exposed to chlorine.

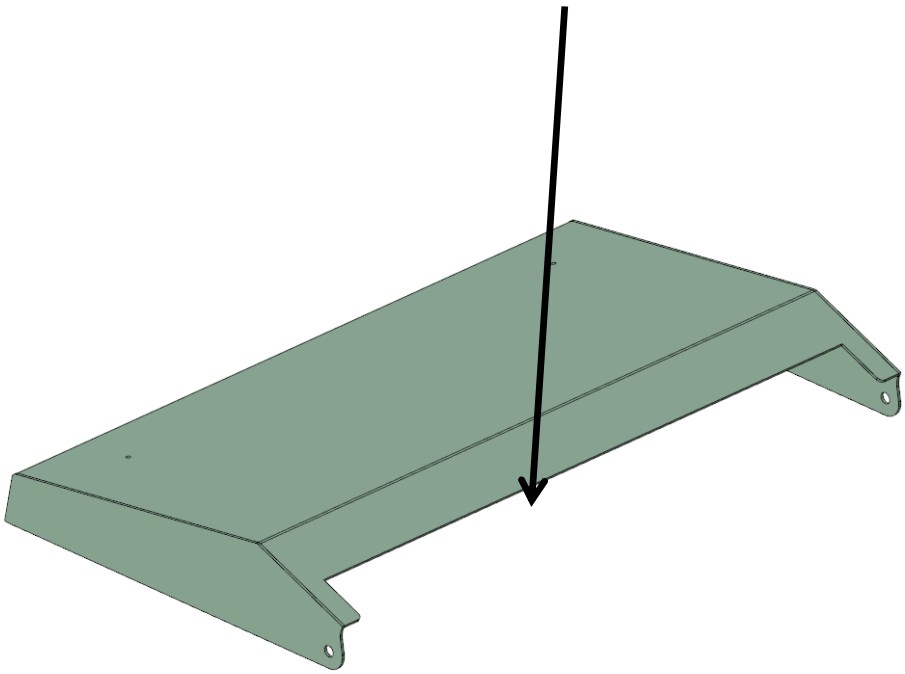
Option 2: painted aluminium performs better than stainless steel against chlorine corrosion. However, aluminium is relatively soft and prone to deformation and abrasion,Once the protective coating is worn away, the substrate will corrode easily.

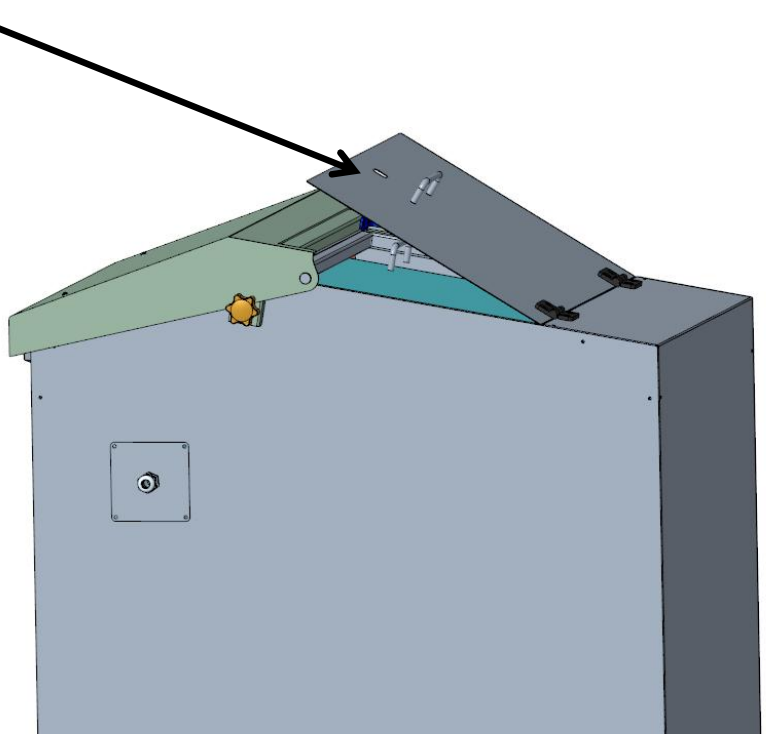
Option 3: PVDF features excellent UV resistance and chemical resistance.

We suggest to choice option 3 that use PVDF material.

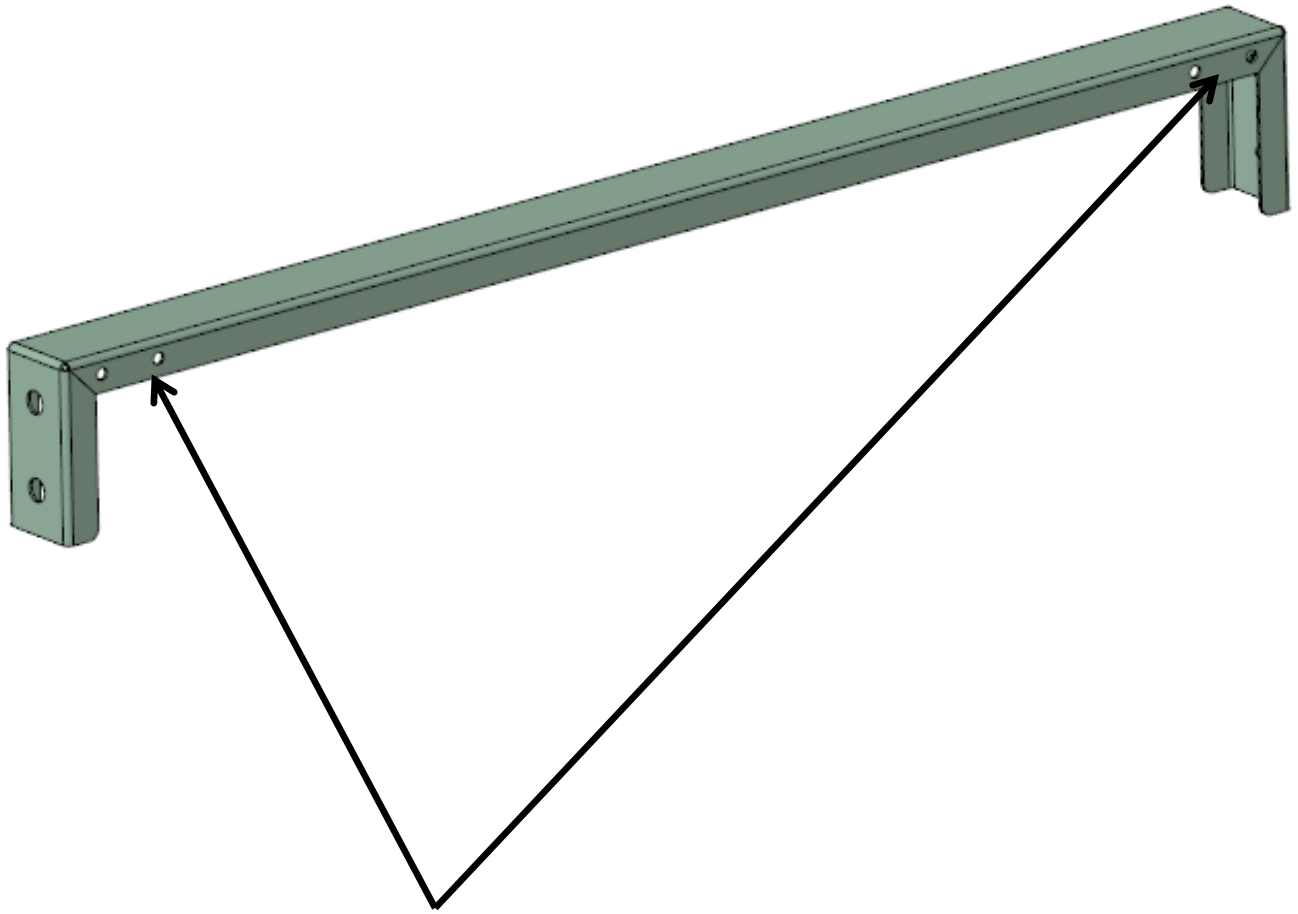
Design Overview--Changed items compare to original design

We need to rework the original design to remove some material from the sun-proof cover in order to open the rear top cover for adding the salt.





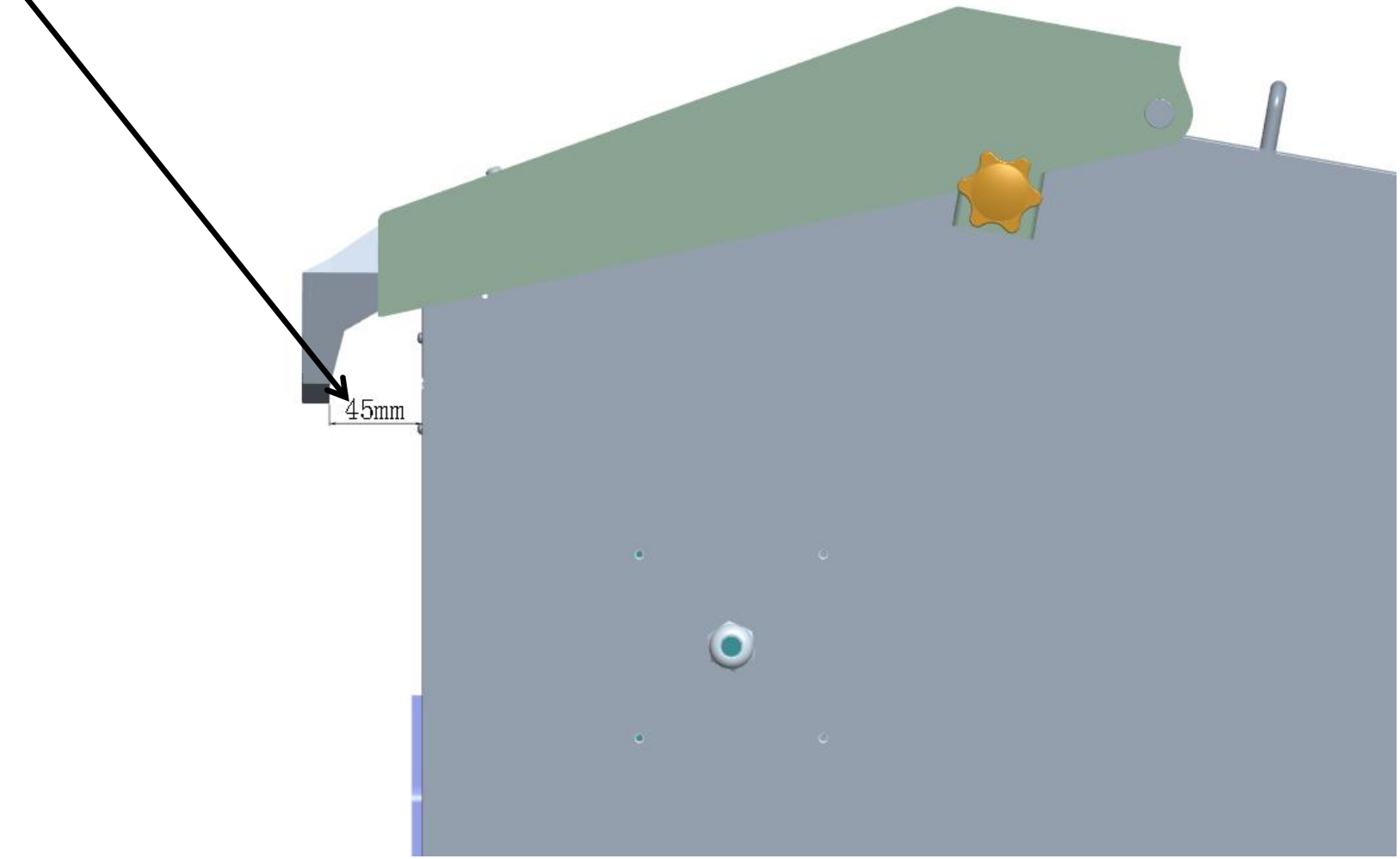
Design Overview--Changed items compare to original design



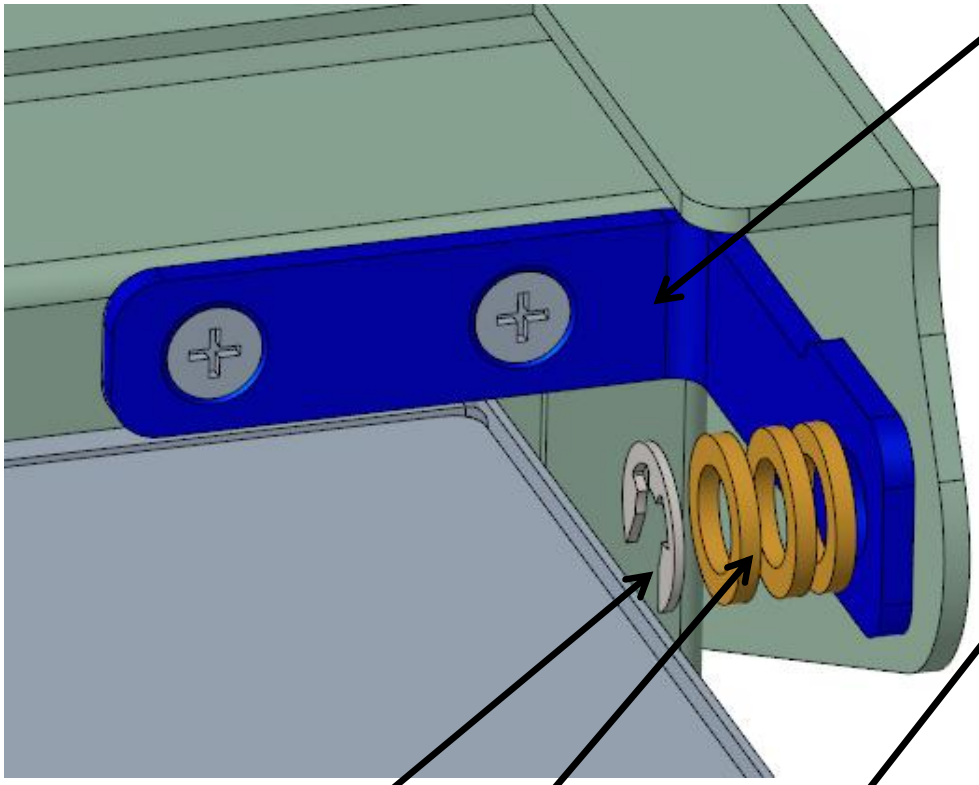
We need to rework： drill four holes in the base bracket to secure the additional bracket.

Design Overview--Changed items compare to original design

The handle will not be far away from the front panel, will not have this gap after rework.

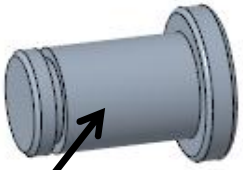


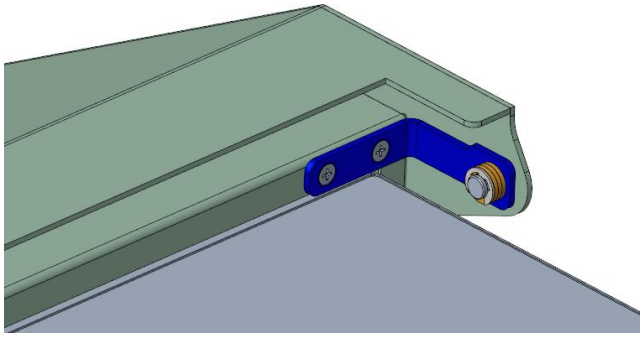
Option 1: Use SUS304 and US 316



3mm thickness SUS316 brackets with painting

Process:Laser cut ➡ Bending ➡ Champer ➡ Powder coated



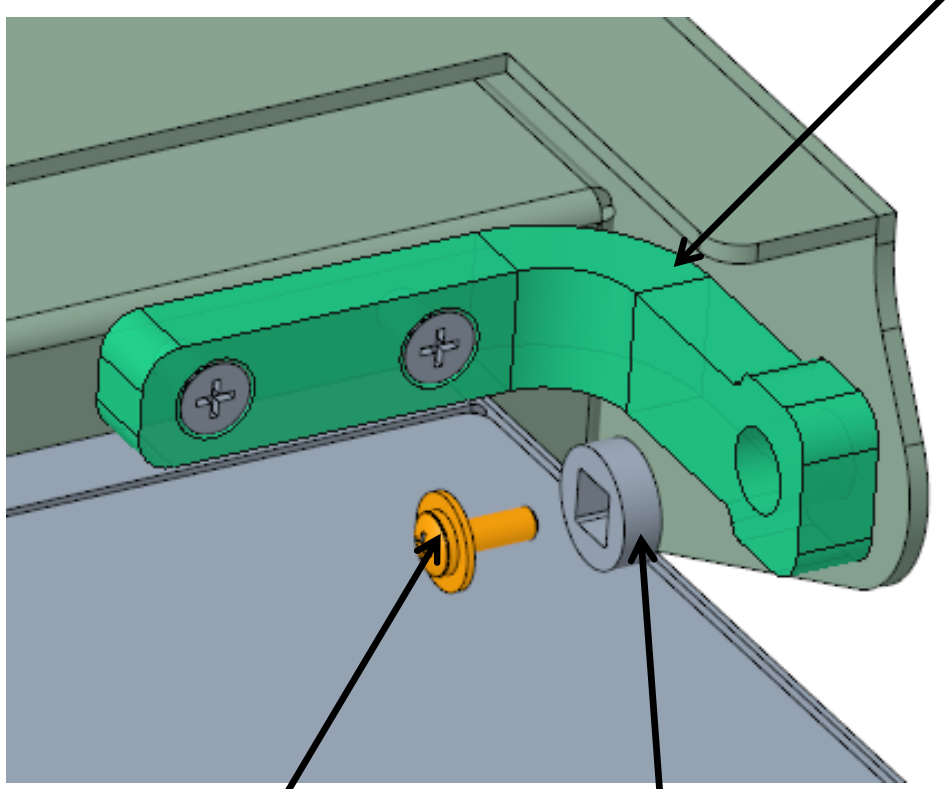


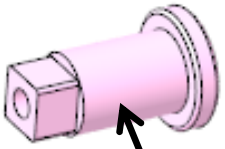
E-ring,washers and diameter 10mm E-ring shaft are SUS304, standard parts,always be fricting,can’t painting

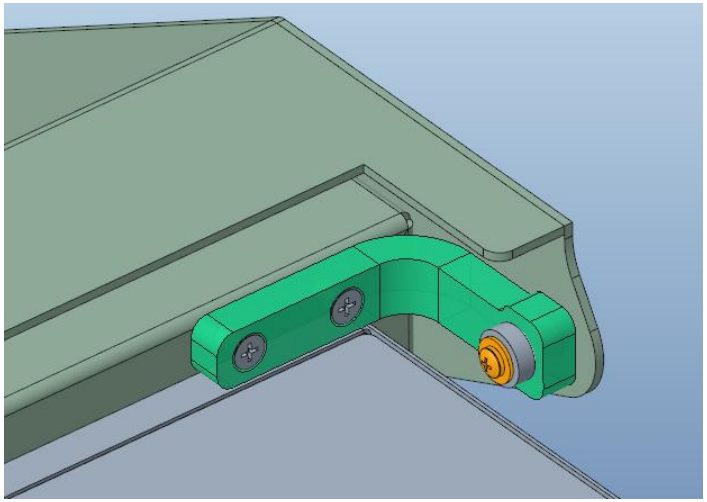
Option 2: Use aluminal

8mm thickness aluminal brackets with anodizing and painting

Process:
CNC ➡ anodizing ➡ Powder coated



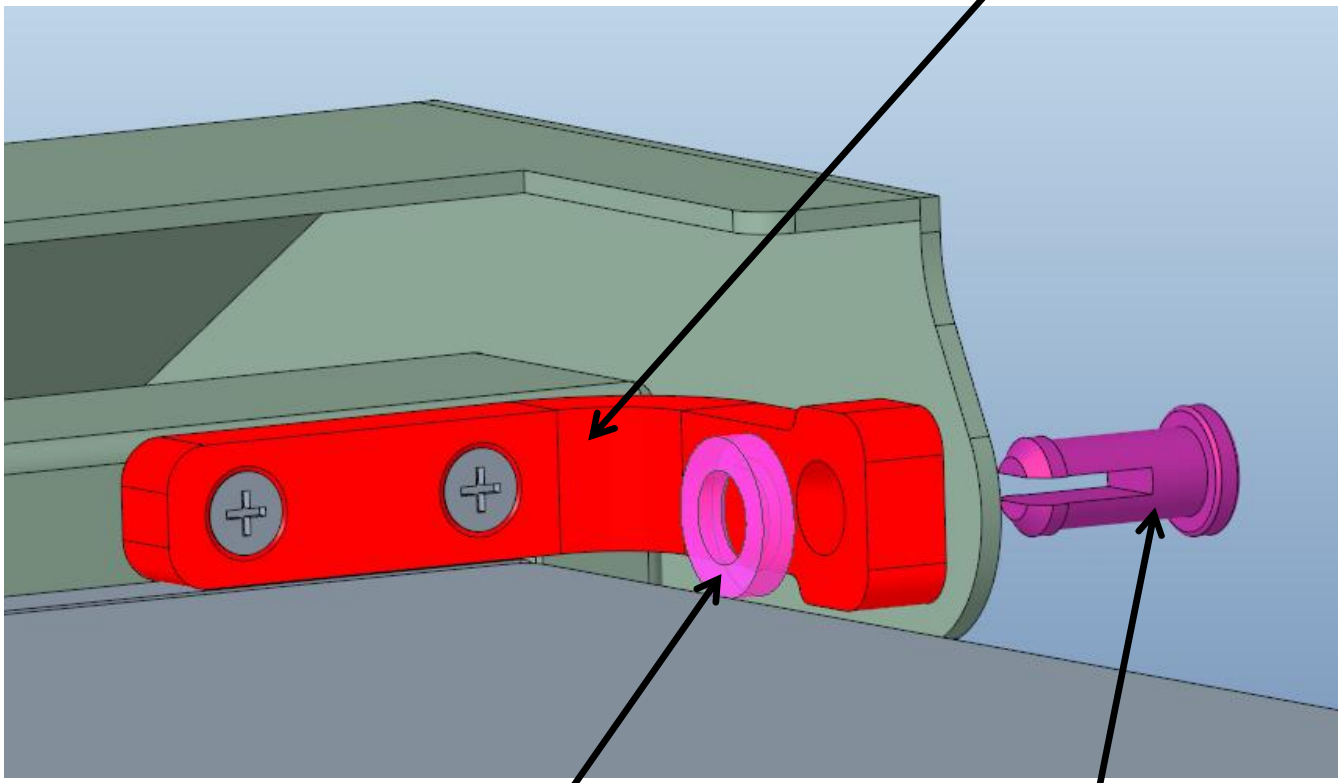


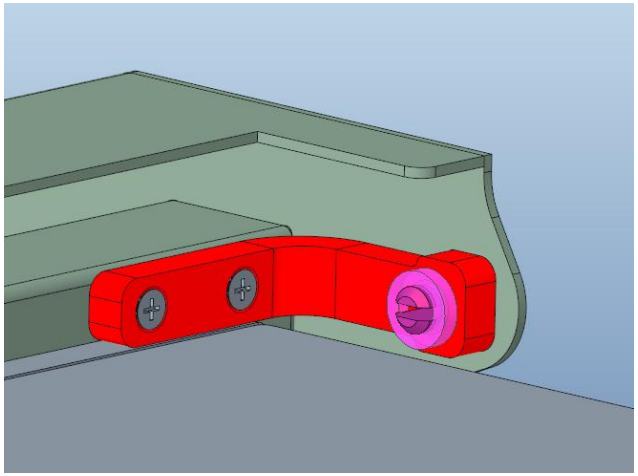


SUS304 PWM4X12 screw ,AL washers and diameter 10mm AL shaft
Process for Al parts:
CNC ➡ anodizing ➡ Powder coated

Option 3: Use PVDF

8mm thickness PVDF brackets
Process: CNC





4mm thickness PVDF washer
Process: CNC

diameter 10mm PVDF shaft